

Alex Mercer

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Professional Summary

Passionate Systems and Low-Level Software Engineer with 4+ years of experience in x86_64 and ARM Assembly programming, C/C++ firmware optimization, and kernel debugging. Proven track record in reverse engineering

Technical Skills

Languages: Assembly (x86, x86_64, ARM), C, C++, Python, Bash

Tools & Debuggers: GDB, WinDbg, IDA Pro, Ghidra, Git, Make, CMake

Operating Systems: Linux Kernel Internals, Windows Internals, FreeRTOS

Domains: Embedded Systems, Reverse Engineering, Performance Optimization, SIMD (AVX2)

Work Experience

Senior Systems Engineer - Embedded Core Inc. (2022 - Present)

- Developed and optimized x86 and ARM Assembly routines for cryptographic algorithms.
- Collaborated with C++ developers to profile high-throughput trading engine using GDB.
- Disassembled and reverse engineered legacy firmware to eliminate security vulnerabilities.

Firmware Developer - Quantum Devices (2020 - 2022)

- Implemented low-level device drivers and bootloaders for ARM Cortex-M microcontrollers.
- Utilized SIMD vectorized instructions to accelerate signal processing by 35%.

Education

B.S. in Computer Science - University of California, Berkeley (2016 - 2020)